Name of the Student: _____Roll no: _____

CLASS: - T.E (CSE) Division: Course: - PCC305CSE- Machine Learning Lab

Experiment No. 01

** Basic Feature Engineering Techniques **

Marks: /10

Date of Performance: / /2026 **Sign with Date:**

Aim: To prepare a clean and model ready dataset for machine learning analysis.

Lab Objective: To apply basic feature engineering techniques.

Problem Statement:

Apply basic feature engineering techniques such as handling missing values, encoding categorical variables, and scaling numerical attributes on the suitable open access dataset to prepare a clean and model ready dataset for machine learning analysis.

Theory:

Dataset Overview

We will use the **Titanic Passenger Survival Dataset**. It contains demographic and travel information for passengers on the Titanic, along with whether they survived.

- **Source:** Hosted publicly on GitHub (via Data Science Dojo).
- **Why this dataset?** It requires significant cleaning before a machine learning model can process it, making it the perfect sandbox for your first assignment.

Prerequisites

Before beginning, ensure you have the following Python libraries installed in your environment:

- pandas (for data manipulation)
- numpy (for numerical operations)
- scikit-learn (for machine learning preprocessing tools)

Questions:

1. What is feature engineering?
2. Differentiate between One-Hot Encoding and Label/Ordinal Encoding
3. Why are missing values handled?
4. Why is feature scaling required?
5. What does StandardScaler do?
6. Implement a program to apply basic feature engineering techniques such as handling missing values, encoding categorical variables, and scaling numerical attributes on the suitable open access dataset to prepare a clean and model ready dataset for machine learning analysis.

Attendance and Timely Submission (10)	Execution (10)	Observations and Records (10)	Journal Documentation (10)	Oral (10)

Signature of Subject Teacher